



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic decisions.

Dataset at a Glance

3,900

Purchases

18

Columns

50

Locations

37

Missing Values

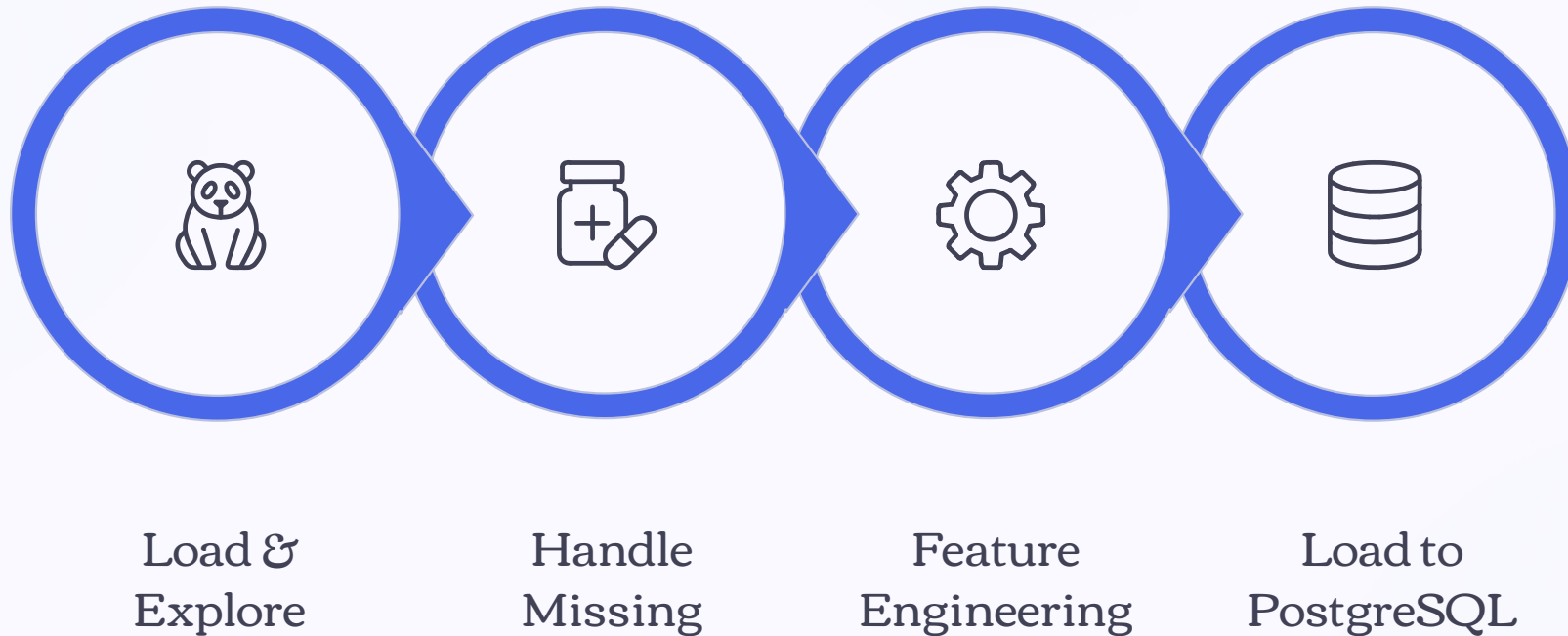
Review Rating column



Key Features

- Demographics: Age, Gender, Location, Subscription
- Purchases: Item, Category, Amount, Season
- Behavior: Discount, Promo Code, Frequency, Rating

Data Preparation in Python



Columns standardized to snake_case; `promo_code_used` dropped as redundant with `discount_applied`.



Revenue by Gender

Male

\$157,890 total revenue

Female

\$75,191 total revenue

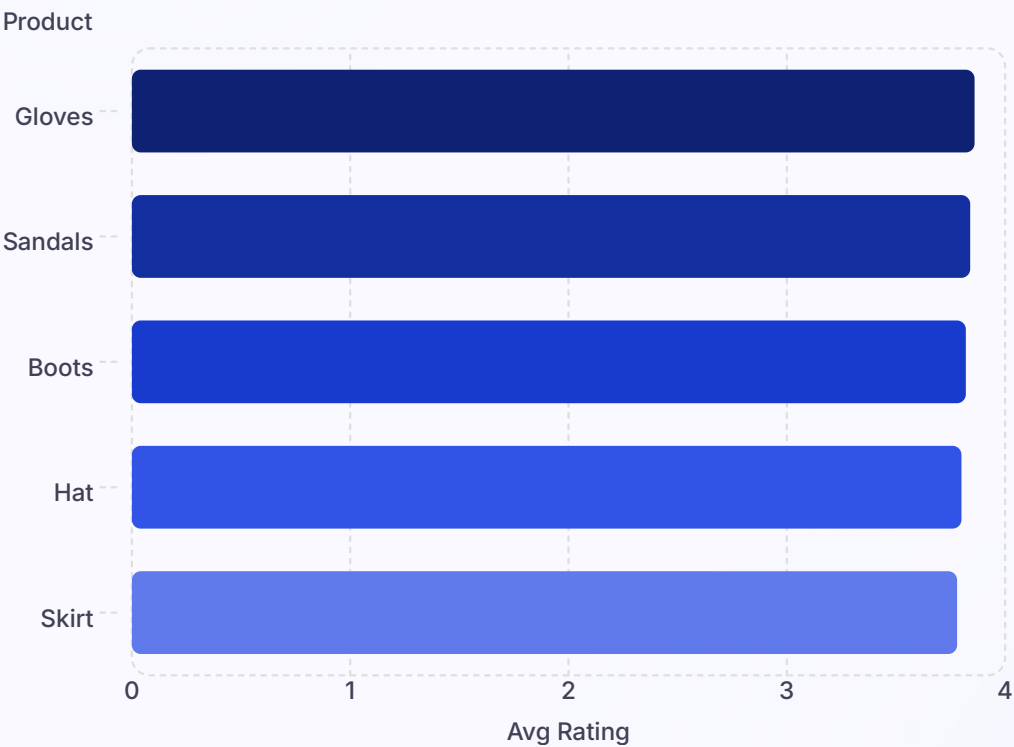
Male customers generated **2.1× more revenue** than female customers across all categories.

Discount Behavior & Top Products

High-Spending Discount Users

839 customers used discounts yet spent *above* average purchase amount.

Top 5 Products by Rating



Gloves

3.86 ★

Sandals

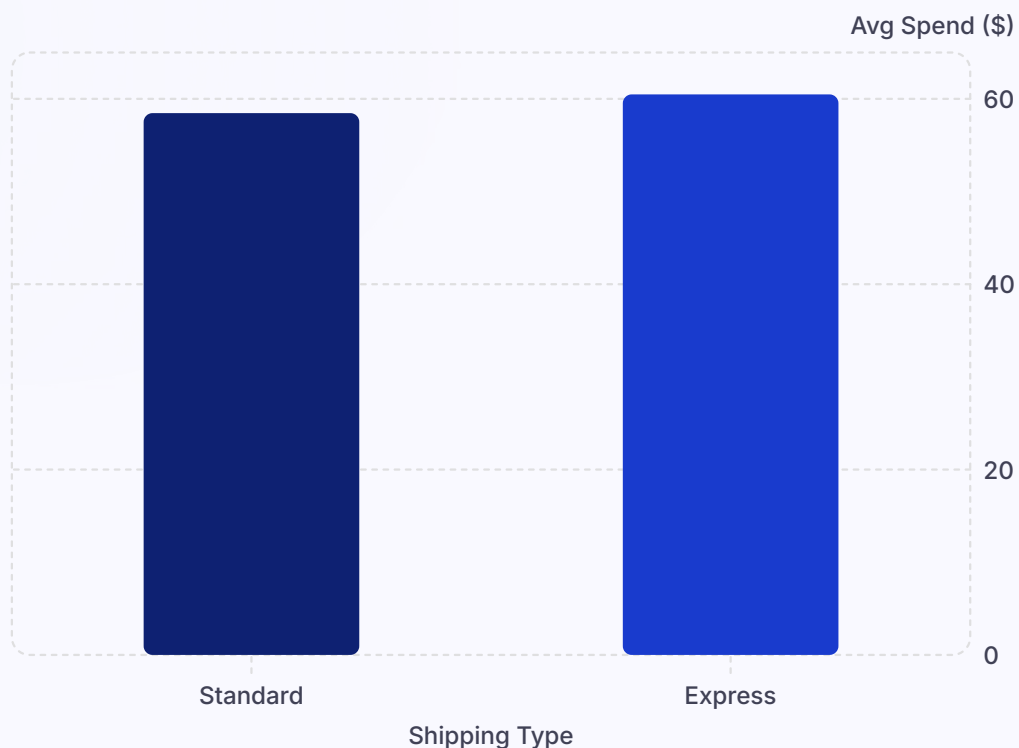
3.84 ★

Boots

3.82 ★

Shipping & Subscription Insights

Avg Spend by Shipping Type



Subscribers vs. Non-Subscribers

Subscribers (27%)

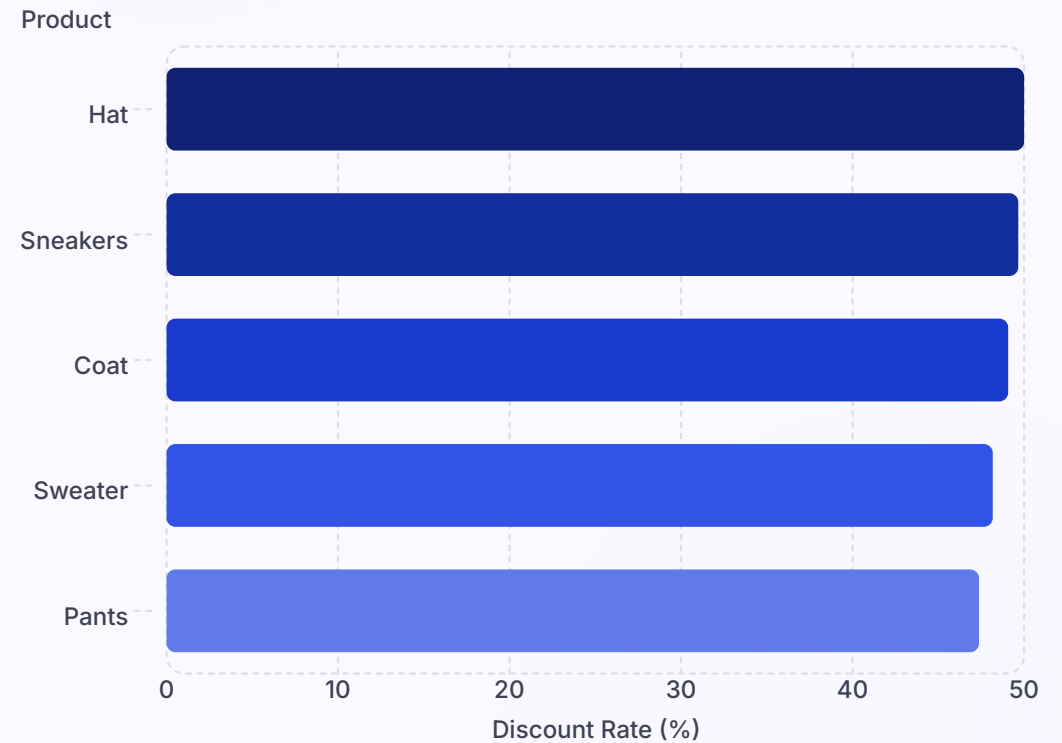
1,053 customers ·
\$62,645 revenue

Non-Subscribers (73%)

2,847 customers ·
\$170,436 revenue

Discount-Dependent Products

Top 5 products with the **highest percentage of discounted purchases**



Customer Segmentation



Loyal

3,116 customers



Returning

701 customers



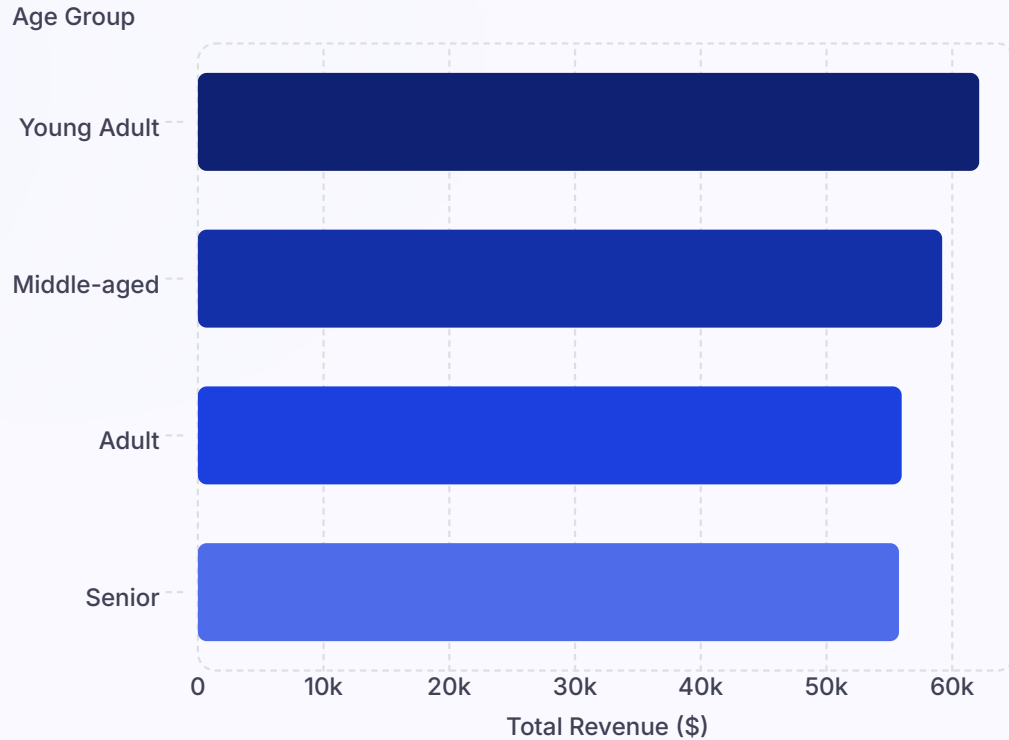
New

83 customers

Repeat buyers (>5 purchases): **958 subscribers** vs. **2,518 non-subscribers**.



Revenue by Age Group



Key Takeaway

Young Adults lead in revenue contribution at **\$62,143**, closely followed by Middle-aged at \$59,197.

i All four age groups show relatively balanced revenue – opportunity to grow each segment.



Promote exclusive subscriber benefits



Reward repeat buyers → Loyal segment



Balance sales boosts with margin control



Focus on high-revenue age groups & express-shipping users